

[Mobile Dev] 16th May: React Navigation Masterclass [1]

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React navigation: A library used for standard navigation used to implement routing logic.

- It helps us move between screens.

Starter code: `bunx create expo-app t blank-typescript@sdk`

cmd: `bun add @react-navigation/native`

Installing dependencies: `bunx expo install react-native-screens react-native-safe-area-context`

Navigators: A thing that controls how screens move and behave in your app.

Types of navigator:

1. Stack Navigator
2. Tab Navigator
3. Drawer Navigator

Static Navigation: Here screens are predefined. We define all screens beforehand.

Dynamic Navigation: Screens are added/rendered conditionally. Changes are based on:
auth data, API data, user roles, feature flags etc.

Steps to follow:

1. `npm add @react-navigation/native`
2. `npm expo install react-native-screens react-native-safe-area-context`
3. `npm add @react-navigation/elements`
`npm add @react-navigation/native-stack`

Code in App.tsx

```
import * as React from 'react';
import { View, Text } from 'react-native';
import { createStackNavigator } from '@react-navigation/stack';
import { createNativeStackNavigator } from '@react-navigation/native-stack';

const Stack = createNativeStackNavigator();

const screens = {
  Home: HomeScreen,
  Details: DetailsScreen,
  Profile: ProfileScreen
};

const Navigation = createStackNavigator(Stack);

export default function App() {
  return <Navigation/>
}
```


In HomeScreen.tsx

3

```
import { StyleSheet, Text, View } from 'react-native'
import React from 'react'
import { Button } from '@react-navigation/elements'
import { Link, useNavigation } from '@react-navigation/native'

const HomeScreen = () => {
  const navigation = useNavigation <any>()
```

return (

<View>

<Text> HomeScreen </Text>

<Button onPress={() => navigation.navigate("details")}>

> Go to Details </Button>

</View>

export default HomeScreen

const styles = StyleSheet.create({})

Types of methods in Stack Navigation

1. **navigate()** → go to a screen by name
2. **goBack()** → go to previous screen
3. **push()** → Always adds a new instance
4. **replace()** → replace current screen / user can't go back to this screen. eg SplashScreen
5. **popToTop()** : go to first screen
6. **pop()** : Remove screen.

Also you can define "Initial Route Name", decides which screens open first.

4

Accept routes

```
const DetailsScreen = ({ route: { args } }) => {
```

```
  const { username } = route.params;
```

```
  return (
```

```
    <View>
```

```
      <Text> {username} </Text>
```

```
      <Button title="Go Profile" onPress={() =>
        navigation.replace('Profile')} />
```

```
    </View>
```

assignment: Now can we send initial values.

Stack Navigator: Here screens are placed on top of each other like a stack.

Why? Screen-to-screen navigation
+ need back button support
+ want push/pop navigation flow

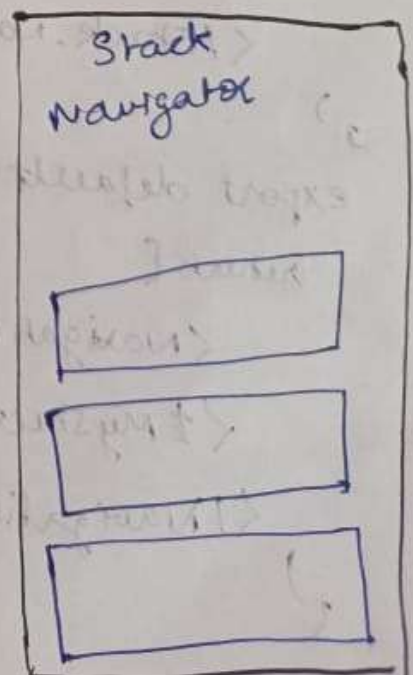
eg: Authentication flow

e-commerce apps

Social media apps

multi-step apps

Cred like apps



Types of stack : [5]

1. Stack

2. Native Stack

Dynamic Stack Navigator :

→ Import * as React from 'react';
import { NavigationContainer } from '@react-navigation/native';

import { createNativeStackNavigator } from
 '@react-navigation/native-stack';

const Stack = createNativeStackNavigator();

fun MyStack() {

return (

<Stack.Navigator>

<Stack.Screen name = 'Home' component = {HomeScreen}/>

</Stack.Navigator>

})

export default fun DynamicStackNavigator() {

return (

<NavigationContainer>

<MyStack/

</NavigationContainer>

})

To customize, use options: ← route name ← is screen
eg: `<Stack.Screen name = 'Home' component = { HomeScreen} options = { { title = "My Home Page" } } />` [6]
} changes headers title

To hide header,
options = { { headerShown: false } }

ScreenOptions: used for globally implement styles.

Screen is given more, meaning "options" on specific screen has higher priority than "screenOptions".

Assignment: Research about animation.

⇒ Tab Navigation: There are two types:

1. Bottom tabs
2. Top tabs

Tabs gives fast switching, better UX,
Tabs give us button to switch to a main section directly

Tabs navigation means showing different screens using tabs, usually at the bottom of the app.

→ Tap on tab to switch screens. eg in Instagram, Youtube etc

cmd: bunx expo install @react-navigation/bottom-tabs

eg:

7

```
return (  
  <Tab.Navigator>  
    <Tab.Screen name = 'name' component = {HomeScreen} />  
  </Tab.Navigator>  
)
```

eg: to customize it,

```
<Tab.Navigator initialRouteName = 'Search' screenOptions =  
  {{ headerShown: true }}>
```

```
<Tab.Screen name = 'Home' component = {HomeScreen}  
  options = {{
```

```
    title: "Dashboard",
```

```
    tabBarLabel: "Start"
```

```
  }}  
</Tab.Screen>
```

```
</Tab.Navigator>
```

will change Home tab
name to "Start"

To add icons. → bunx expo install @expo/vector-icons

Top Tabs: eg in Instagram, chats to feed etc.

Drawer Navigation: It is the side menu navigation pattern.

eg when you swipe from left, or tap on hamburger icon
☰, a menu slides in.

cmd: `npm install @react-navigation/drawer`

8

Some expo dependencies:

- `react-native-gesture-handler`
- `react-native-safe-area-context`
- etc.

Assignment: Build a Food Delivery App UI
A complete multi-screen React Native App
using all navigation patterns

